

LPDC comma alignment and phase matching on the CTS/KR260 GTHE4: what we tried and why it was reverted

Date: 2026-08-14 **Status:** SUPERSEDED — see addendum **Target:** CTS / KR260, Xilinx GTHE4 (Kria K26), White Rabbit

Addendum (2026-08-19). The root-cause conclusion of this note (§6: the RXPI direct-tag is inherently non-deterministic per relink; sub-ns needs a DMTD clock) is **disproven**. The scatter had three concrete causes – an uncalibrated master-side timestamp phase (the sweep never ran on a master), an uncompensated per-relink rxslide UI latency, and a GTHE4 mod-10 wrap in that latency. With those fixed the direct-tag path delivers per-relink PPS deterministic to ± 50 ps, no DMTD. See [lpdc-rxpi-deterministic-pps.md](#). The bring-up record below (the seven walls) remains valid and was the foundation of the successful second campaign.

Scope. This note records an attempt to add LPDC (Low Phase Drift Calibration) deterministic RX comma alignment on top of the existing RXPI fine-phase path on the CTS GTHE4 transceiver, the sequence of issues found during bring-up and how each was solved, and the point at which the effort stopped. The link-layer determinism goal was met; the sub-nanosecond *phase* goal was not, for a reason that sits one layer below LPDC. The work was reverted; two useful parallel fixes were kept. This documents the outcome so the direction is not blindly re-attempted.

1. Goal

Combine two deterministic-latency strategies on the CTS GTHE4:

- **LPDC** — bypass the RX elastic buffer and align the 8b10b comma to a *fixed* tap, so the RX byte/UI latency is deterministic (the classic White Rabbit “low phase drift” method).
- **RXPI** — the existing CTS mechanism that reads the CDR phase-interpolator (`dmonitorout`) and sweeps an MMCM to measure/compensate the sub-UI phase.

Target outcome: a robust WR link with **reproducible, sub-ns phase across relinks and power-cycles**.

2. What we built (the LPDC integration)

RX-only conversion; the TX path was left exactly as the working RXPI design.

LAYER	CHANGE
GT IP (<code>gthe4_phy.tcl</code>)	RX elastic buffer bypass + RAW 20-bit , internal 8b10b / comma alignment off; QPLL0 fractional-N, <code>dmonitor</code> , <code>txpippm</code> preserved
Fabric datapath (<code>rxpi_lp_adapter.vhd</code> , new)	raw 20-bit → <code>gtx_comma_detect_lp</code> (fixed-tap barrel shift) → fabric <code>gc_dec_8b10b</code> ×2; instantiates <code>lpdc_mdio_regs</code>
Register access	endpoint <code>phy_mdio_master</code> wired to <code>lpdc_mdio_regs</code> (MDIO PHY-specific window)
Fine phase	RXPI <code>dmonitor</code> sweep kept unchanged (<code>xwrc_gthe4_rxpi.vhd</code>)
Firmware (<code>lpdc_gthe4_rxpi.c</code>)	merged an LPDC comma-target FSM with the existing sweep FSM

3. Bring-up: the walls and the fixes

Each item below was a distinct failure found on hardware and resolved before the next surfaced.

#	SYMPTOM	ROOT CAUSE	FIX
1	Slave never re-rolls, deadlocks	RX-retry gated on <code>phy_ready</code> , which needs <code>buffbypass_done</code> ; wrong tap + stuck <code>buffbypass</code> = deadlock	evaluate the comma detector directly, re-roll immediately, add a timeout
2	Link flaps / rejects a good tap	<code>comma_pos_valid</code> (asserts only during idles) was gating acceptance	accept on the latched <code>aligned</code> bit instead
3	Comma position jitters, no sync	RX buffer-bypass alignment started before the CDR relocked	gate the <code>buffbypass</code> reset on <code>rx_reset_done</code> ; reset the comma detector on <code>rx_rst_done</code>
4	“Mostly fails to find the comma”	the comma lands on a single parity per boot, and the parity flips per power-cycle (the long-standing 98%-odd bitslide)	adaptive parity target in firmware
5	PHY aligned but <code>wr0</code> down	auto-negotiation was kicked once at boot, <i>before</i> the PHY aligned, and never restarted	restart AN on alignment
6	AN never completes (<code>lpa=0000</code>)	on an odd target tap, the comma barrel-shift’s 1-bit borrow corrupts the /C1//C2/ D21.5/D2.2 marker (the comma and config bytes decoded fine); an even tap has no borrow	captured the decoded RX words (<code>rxpi pat</code>) to prove it; steer to an even tap
7	Reaching an even tap on every boot	PMA <code>rxslide</code> is rejected by the IP in the bypass/RAW config (only <code>PCS / OFF</code> allowed)	PCS <code>rxslide</code> + a fabric FSM that bit-slips the comma to tap 0 deterministically

Diagnostic tooling built along the way (all in the now-reverted driver): `rxpi stat` (PHY + endpoint sync/AN/LPA), `rxpi pat` (latch & dump decoded RX words via `lpdc_mdio_regs.idle_pat`), `rxpi an <0|1>`, `rxpi ld` (tune `phase_ld`), `rxpi fixoff` (skip the sweep, pin a fixed offset).

4. The working result

After the PCS-rxslide-to-tap-0 fix, the **link layer was fully deterministic**:

- Both ends align at tap 0 every boot: `pos=0 aligned=1`.
- Auto-negotiation completes: `link=1 aneg=1 lpa=4020`.
- Slave locks: `MPL1 lck:1`.
- **Fiber replug re-links cleanly**, regardless of session parity.

This confirmed the LPDC comma alignment did its job: the byte/UI-level latency was made deterministic and the WR Ethernet link was solid and reproducible.

5. Phase matching — where it stopped

With the link working, the **per-relink phase offset stayed ~2–8 ns and was not reproducible** (the spread was under 16 ns, i.e. within one 62.5 MHz period). Two coupled problems:

- **Intermittent lock loss (~every 3 min)**. A CDR-PI rollover spike trips `phase_ld` → unlock → PTP desequences (`slave_handle_followup`) → the FSM **re-runs the sweep**, which **jumps the offset mid-session**.
- **The per-relink sweep delta is not reproducible**. Live dumps showed `delta` swinging ~12 → 179 (×200 ps) with the sweep's `±56/4` boundary heuristic flipping and a coarse-cycle ambiguity in the coarse/fine stitch.

The decisive experiment was `rxpi fixoff`: **skip the sweep entirely and pin a fixed ptracker offset**. The offset **still scattered across relinks**. That rules the sweep out as the root cause — the underlying reference is moving.

6. Root cause

The RXPI **direct-tag** phase measurement (the accumulated CDR phase-interpolator reading, fed to the SoftPLL as `direct_tag0`) references the **random CDR lock phase**, not a local clock. So:

- The comma alignment pins the **UI** (byte boundary) — deterministic.
- Nothing in this scheme pins the **sub-UI recovered-clock phase** reproducibly — the direct-tag carries a per-relink offset it cannot resolve, and the sweep was trying to correct a moving target (so “measure once and store” was never going to work).

CTS uses the direct-tag path precisely **because the board has no DMTD clock** (`clk_dmtd` is tied to `0`, `g_direct_tag = true`). A DMTD clock references the **local** oscillator, giving a phase that is deterministic modulo one period — which is exactly what a reproducible sub-ns measurement needs.

Frequency **syntonization** was never the problem: the boards held ~9 ps RMS throughout.

7. Conclusion and path forward

- **LPDC succeeded at its layer.** Deterministic byte/UI comma alignment (tap 0 via PCS rxslide) produced a robust, reproducibly re-linking WR connection with correct auto-negotiation. That is a genuine, kept-worthy result.
 - **Sub-ns phase is blocked one layer below LPDC**, in the RXPI direct-tag reference. No amount of comma alignment or sweep bookkeeping fixes a phase reference tied to the random CDR lock.
 - **The real path to <1 ns is not more datapath work — it is a clock.** Provide a **DMTD clock** (a few ppm off the WR reference, e.g. derived from the on-board Si5344) and switch the ptracker from direct-tag to the **standard WR DDMTD**. That is a board/clocking change.
 - **Decision: reverted** the LPDC work in both `wr-cores` and `wrpc-sw`, back to pure-RXPI. Until a DMTD clock is available, pure-RXPI is the working baseline — fine for frequency / syntonization use.
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8. Repository state after revert

Reverted (my LPDC changes): `board/gthe4-rxpi/gthe4_phy.tcl`, `board/gthe4-rxpi/xwrc_board_gthe4_rxpi.vhd`, `platform/xilinx/UltraScalePlus/GTHE4-rxpi/Manifest.py`, `platform/xilinx/UltraScalePlus/GTHE4-rxpi/rxpi_lp_adapter.vhd` (deleted), and `wrpc-sw/dev/lpdc_gthe4_rxpi.c`.

Kept (two parallel, non-LPDC improvements made during the session — do not discard):

FILE	WHAT IT DOES
platform/xilinx/UltraScalePlus/GTHE4-rxpi/xwrc_gthe4_rxpi.vhd	b_rxpi continuous-unwrap CDR accumulator — kills the half-UI rollover spikes on the pure-RXPI path
wrpc-sw/softpll/spll_main.c	looser phase_ld (threshold 2000 / delock 8192)

Note. The generated gthe4_phy IP on disk is still the LPDC (bypass/RAW) version from the last resynth. Re-run the reverted gthe4_phy.tcl to regenerate the original (buffered, 8b10b, 16-bit) IP before the next build, or the reverted board component/port map will mismatch.